

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 27

DECIMAL TO HEXADECIMAL

AIM: To write and execute an 8085 assembly language program to convert decimal to hexadecimal number and store result in memory.

ALGORITHM

1. Store the decimal number at memory location **2050H**.
2. Load the decimal number into the accumulator.
3. Divide the number by **16** repeatedly.
4. Store each remainder as a hexadecimal digit.
5. Continue until the quotient becomes zero.
6. Arrange the remainders in reverse order to get the hexadecimal number.
7. Store the hexadecimal result in memory.
8. Stop the program.

CODE:

LDA 2050H

MVI B,00H

LOOP: SUI 10H

JC STORE

INR B

JMP LOOP

STORE: ADI 10H

MOV C,A

MOV A,B

ANI 0FH

RLC

RLC
RLC
RLC
ORA C

STA 2051H
HLT

OUTPUT:

Memory

0x0 - 0x1B

0x2050

+

Displaying Memory Locations from 0x2050 to 0x2050

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x2050

1A

1 MVI A,26H

2 MOV B,A

3

4 ANI 0FH

5 MOV C,A

6

7 MOV A,B

8 ANI 0F0H

9 RRC

10 RRC

11 RRC

12 RRC

13

14 MOV D,A

15 ADD A

16 ADD A

17 ADD A

18 ADD D

19 ADD D

20 ADD C

21

22 STA 2050H

23 HLT